

Measurement Validity & Instrument Reliability Analysis

1) Student Information

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Course: 3

Research Topic: CS2 Match Outcome Prediction Using Machine Learning

2) Instrument Overview

This study utilizes a secondary dataset aggregated from publicly available esports platforms, primarily HLTV.org. The instrument consists of four key variables: one binary dependent variable (Match Outcome) and three independent variables (Team Combat Statistics, Head-to-Head Match Context, and Map Strength). Because the data is numerical and derived from past match performances, we rely on a mix of binary, interval, and ratio scales. Using a secondary dataset is highly appropriate for this research question because machine learning predictions require massive amounts of objective, standardized, and historical data, which cannot be practically gathered through self-designed surveys or primary experiments.

3) Construct Validity

Construct validity ensures that our variables accurately reflect our theoretical framework.

- **Team Combat Statistics (X1):** We defined this conceptually as a team's fighting ability. We operationalize this using the mean HLTV 2.0 rating and mean Kills Per Round (KPR) over the past 3 months. These metrics validly represent combat strength because they aggregate individual player mechanical skill and engagement success into a measurable team output.
- **Head-to-Head Context (X2):** Conceptually, this represents the historical and psychological advantage a team has over a specific opponent. We operationalize this via the direct matchup win rate over the past 12 months. This aligns well with our theoretical construct because beating a rival repeatedly indicates both strategic familiarity and a psychological edge.
- **Map Strength (X3):** This measures a team's historical comfort on a specific map. By operationalizing this as the map-specific win rate over the past 3 months, we directly capture a team's specialized tactical preparation and utility execution on that specific terrain.

4) Reliability Analysis

Because this study relies entirely on secondary historical data, utilizing internal consistency metrics like Cronbach's alpha is not applicable. Instead, the reliability of our instrument is rooted in source credibility and measurement consistency.

HLTV.org is the industry standard for Counter-Strike data, calculating metrics like the 2.0 rating using strict, standardized algorithmic formulas applied universally across all professional matches. Furthermore, by framing our variables as 3-month and 12-month rolling averages rather than looking at isolated games, we ensure temporal stability and smooth out random performance spikes. However, reliability may still be slightly compromised in instances where a team has played very few matches within the 3-month window, creating a small sample size that makes their average statistics more susceptible to statistical noise.

5) Scale Justification

- **Match Outcome (Y):** We use a binary scale (1 for a win, 0 for a loss). This is the standard requirement for training classification models like Logistic Regression and Random Forest.
- **Team Combat Statistics (X1):** KPR is a true ratio scale, while the HLTV 2.0 rating functions as an interval scale. These continuous scales are necessary to capture the nuanced gradients in team skill levels.
- **Head-to-Head (X2) and Map Strength (X3):** Both are measured as percentage win rates, which are ratio scales. This allows the models to mathematically weight exactly how dominant a team is in certain situations. While tree-based models (Random Forest, XGBoost) do not require normalization for these scales, we will need to standardize these continuous variables before feeding them into our Logistic Regression baseline to ensure the coefficients are interpretable.

6) Measurement Limitations

Drawing on our prior ethical risk assessment, our measurement approach has notable limitations. First, the dataset suffers from regional data bias, heavily overrepresenting Tier-1 European and North American teams. Consequently, the model's measurements for teams from the CIS or APAC regions will be inherently noisier and less reliable. Secondly, our quantitative instrument cannot measure critical unobserved variables, such as internal team communication, mid-round calling, and player mental states under pressure. Finally, because we rely on fixed 3-month and 12-month averages, sudden roster changes or abrupt shifts in a player's form leading up to match day are not captured by the instrument, potentially skewing predictions.

7) Improvement Plan

To strengthen the instrument's validity in the future, we could expand our independent variables to include more granular metrics, such as clutch success rates (1vX scenarios) or round economy management, which represent tactical depth better than raw kills. Furthermore, transitioning from strict 3-month averages to a weighted moving average - where matches played in the last two weeks are weighted more heavily than matches played two months ago - would help the instrument adapt more reliably to short-term volatility in team form. Finally, utilizing stratified data balancing techniques could help mitigate the regional measurement bias.